

GURUVISHNU NALLAMADUGU

Starkville, MS (Willing to Relocate) | guruvishnu1922@gmail.com | [LinkedIn](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Manufacturing engineer with nearly 3 years of industry experience in tooling and fixture design, DFM/DFA, and assembly process improvement. Cut cold room setup time by 25% across 42 installations at Bharat Electronics and monthly production time by 8% at Tata Sikorsky. Designs in Creo, SolidWorks, and Siemens NX. M.S. in Aerospace Engineering.

TECHNICAL SKILLS

CAD: Creo, SolidWorks, Siemens NX, AutoCAD, CATIA V5

Design & Drafting: GD&T (ASME Y14.5), Tolerance Stack-Up, 2D Engineering Drawings, Sheet Metal and Weldment Design

Analysis: ANSYS, Engineering Hand Calculations, MATLAB, Python

Manufacturing: DFM/DFA, Tooling and Fixture Design, 3D Printing / Prototyping, Time Studies, Lean (5S, Kaizen)

Quality & Documentation: Root Cause Analysis (5-Why), DFMEA, First-Article Inspection, Bill of Materials (BOM), ECO/ECN, SOPs and Work Instructions, Microsoft Excel

PROFESSIONAL EXPERIENCE

Mechanical Design Engineer | Mississippi State University | Starkville, MS Jan 2026 – Present

- Designed and built 40+ custom products for campus departments in Creo, SolidWorks, and Siemens NX.
- Designed a 3D-printed 6-in-1 field instrument (sensors, battery, fan) that replaced 6 separate instruments.
- Redesigned sample dryer racks: fixed warped frames, upgraded the mesh, and added insulated handles.
- Designed wall-mounted storage with adjustable dual-arm clamps that hold irregular tools off the floor.
- Test and validate each design with the requesting department before final build.
- Review designs with manufacturing and product development teams before fabrication.

Graduate Teaching Assistant | Mississippi State University | Starkville, MS Jan 2024 – Dec 2025

- Taught 2 weekly Mechanics of Materials recitations (50+ students each; 110–120 per semester) over 4 semesters.
- Held weekly office hours to re-explain missed concepts and help students one-on-one.

CAD Designer | Tata Sikorsky | Hyderabad, India Dec 2021 – Dec 2022

- Designed flight-control cable pulleys and guide brackets to new dimensions in Creo and SolidWorks.
- Added edge radii to pulleys and brackets, removing manual deburring and cutting monthly production time 8%.
- Updated legacy designs in the company library with edge radii so future designs started from the improved version.
- Created manufacturing drawings with GD&T for machined, cast, and sheet metal parts.
- Ran tolerance stack-ups to catch fit problems before release to production.

Manufacturing Engineer | Bharat Electronics Limited (BEL) | Machilipatnam, India Dec 2020 – Nov 2021

- Built 42 walk-in cold rooms: made insulated wall, ceiling, and door panels to each site's dimensions, assembled evaporator and condensing units from sister-plant parts, and installed at client sites.
- Cut on-site setup time per cold room by 25%, from 4 days to 3, by finding an assembly bottleneck and coordinating with local civil engineers on site.
- Handled service calls after installation, troubleshooting and repairing panels, doors, evaporators, and condensing units.
- Fabricated and installed out-of-stock components on site with the team, keeping installations on schedule.
- Used root cause analysis and 5-Why to fix recurring assembly defects.

PROJECTS

Asteroid Orbit Optimization (Research Project) May 2024 – Present

- Built a Python tool that models and plots asteroid trajectories using celestial mechanics.
- Used NASA JPL Small-Body Database orbital data to find the lowest delta-V needed to move candidate asteroids into stable orbits near Earth, the Moon, or Mars.

EDUCATION

Master of Science in Aerospace Engineering | Mississippi State University | GPA: 3.5/4.0 Jan 2023 – Dec 2025

Bachelor of Science in Aerospace Engineering | Chandigarh University | GPA: 3.0/4.0 Jun 2018 – May 2022